Lecture 38: The Energy Implications of AI (Thermodynamics, Siting, and Grid Reality)

CBE 30235: Introduction to Nuclear Engineering — D. T. Leighton

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Introduction: The AI Grid Collision

The transition to generative AI is fundamentally a thermodynamic problem. Training and running Large Language Models (LLMs) requires massive power density. We are currently witnessing a historic collision between exponential computing demand and the baseload realities of the Midwestern electrical grid.

1 The Local Demand Shock: Northern Indiana

Northern Indiana has quietly become a global epicenter for hyperscale data center development due to land availability, water access, and proximity to major fiber optic routes and the PJM/MISO interconnections.

- **Amazon (AWS):** Developing a massive $11 billion campus in **New Carlisle** and a staggering 2.4 GW secondary site just to the west in **Hobart**.
- **Takanock:** Planning a 1.5 GW generation and data center island in **Goshen**.
- **Google:** Advancing a major 1.2 GW site in **Fort Wayne** (Project Zodiac).
- **Microsoft:** Building a multi-phase, gigawatt-scale campus in **Granger**.

While current operational loads are relatively small, the 2030 forecast for these combined regional campuses represents a continuous demand that fundamentally threatens baseload stability.

2 The Energy Balance: Retiring the Baseload

To understand the crisis, we must look at the U.S. Energy Information Administration (EIA) data for Indiana's electrical profile (2023 baseline).

2.1 Indiana Grid Baseline (10^{12} BTU/yr to GW)

Applying standard thermodynamic conversions (1 BTU $\approx$ 1055.06 J):

$$P(\text{GW}) = \frac{\text{Energy (BTU/yr)} \times 1055.06 \text{ J/BTU}}{31,536,000 \text{ s/yr} \times 10^9 \text{ W/GW}}$$

Energy Source	Grid Share (%)	Average Power (GW)
Coal	59.4%	15.53
Natural Gas	35.3%	9.23
Renewables	5.2%	1.36
Nuclear	0.0%	0.00
Total In-State Gen.	**100.0%**	**26.14**
State Imports	–	5.09
Total Consumption	–	**31.23**

Table 1: Indiana continuous electrical generation and consumption baseline.

2.2 The Hyperscale Demand Forecast (2026–2035)

While the current data center footprint in Indiana is relatively small, the committed hyperscale projects represent an unprecedented step-function in electrical demand. AI training clusters (using advanced GPUs) are extremely power-dense, requiring upwards of 50–100 kW per server rack compared to the 5–10 kW typical of traditional cloud storage.

When we isolate just the immediate Northern Indiana corridor, the projected continuous load from fully engineered hyperscale campuses is staggering. Note that these are the projected values - not all will necessarily reach full build out, although the Amazon New Carlisle project is already at 500MW, the Google project in Fort Wayne is partially operational and the land for the Microsoft project in Granger has been purchased and construction is slated for April/May (although there is community resistance). The Hobart Amazon project is navigating legal challenges.

Company / Developer	Location	Full Capacity Load (MW)	Equivalent Homes
Amazon (AWS)	Hobart	$2,400$	$\sim 1,600,000$
Amazon (AWS)	New Carlisle	$\approx 2,250$	$\sim 1,500,000$
Takanock	Goshen	$1,500$	$\sim 1,000,000$
Google (Project Zodiac)	Fort Wayne	$1,200$	$\sim 800,000$
Microsoft	Granger (St. Joseph Co.)	$1,000$	$\sim 660,000$
Northern IN Total		$\approx$ **8.35 GW**	$\sim$ **5.56 Million**

Table 2: Projected full-capacity continuous electrical load from confirmed hyperscale AI campuses strictly in Northern Indiana (Cleanview Database, 2026).

The Scale of the Crisis

To put this $\approx$ 8.35 GW pipeline into perspective, we look to Northern Virginia (specifically Loudoun County), which is historically the largest data center market in the world, drawing approximately **3.5 to 4.0 GW** of continuous power.

The Northern Indiana corridor alone is attempting to construct a digital infrastructure **approaching or potentially exceeding current Loudoun County operational levels** from scratch over the next decade. This localized, regional pipeline will require **an additional 25% - 30% increase** in the entire electrical consumption of the state of Indiana.

This hyper-aggressive timeline turns a grid deficit into an immediate baseload crisis. To power just these five Northern Indiana facilities using zero-carbon baseload, the region would need to build roughly 8 standard commercial gigawatt-scale nuclear reactors, or over 100 advanced SMR modules (at 80 MW each), all before 2035.

2.3 The Shortfall

Right as AI demand is skyrocketing, our primary baseload is retiring:

- **Retirements:** The massive **Rockport Generating Station** (2.6 GW of coal) is slated to retire its units by 2028.
- **The Make-Up:** The regional grid will see the reactivation of the **Palisades Nuclear Plant** in Michigan ($\approx$ 800 MW). To replace the retiring Rockport plant, Indiana Michigan Power (AEP) has partnered with GE Hitachi, targeting the deployment of **two BWRX-300 SMR units** (totaling roughly 600 MW) at the Rockport site in the mid-2030s.
- **The Reality:** Even assuming Palisades comes back online and both BWRX-300 units are successfully built at Rockport (providing roughly 1.4 GW of new baseload), this covers barely half of the retiring coal capacity, let alone the 8 GW of new AI demand. We are running a massive deficit.

3 The Immediate Bridge: Natural Gas

Because nuclear deployment (even SMRs) is fundamentally slow, the immediate, actionable solution for powering Northern Indiana's AI boom is natural gas. The region has a massive geographic advantage: it sits directly on major high-pressure interstate natural gas pipelines. Two distinct engineering and business models are currently exploiting this infrastructure to bridge the gap.

3.1 Model A: The Co-Located Utility (St. Joseph Energy Center)

Located in **New Carlisle**, the St. Joseph Energy Center is a state-of-the-art Combined Cycle Gas Turbine (CCGT) plant operating immediately adjacent to the planned Amazon (AWS) mega-campus.

- **Current Capacity (Phase I):** Currently operational, producing approximately 700 MW of continuous baseload.
- **Expansion (Phase II):** An active expansion is underway to add a second block of roughly 700 MW, bringing the total site capacity to $\approx$ 1.4 GW.
- **The AWS Synergy:** By purchasing land directly next door to the plant, Amazon is positioning itself to draw immense power with minimal transmission loss or grid congestion. Agreements between the developers, AWS, and the local utility (AEP/I&M) effectively dedicate a massive portion of this expanding fossil-fuel capacity to the data center, shielding the tech giant from broader grid instability while they wait for 2030s nuclear options.

3.2 Model B: The "Behind-the-Meter" Island (Takanock in Goshen)

While hyperscalers like Microsoft and Google are highly vocal about a future transition to nuclear power, infrastructure developers like **Takanock** are operating on a purely fossil-fueled, zero-wait strategy.

- **The Scale:** A planned 1.5 GW data center campus in **Goshen**.
- **Simultaneous Construction:** Instead of waiting a decade for the regional grid (PJM) to upgrade substations, string new high-voltage transmission lines, or approve SMR permits, Takanock is building a utility-scale natural gas power plant *on the exact same property* at the same time as the data center.
- **The Engineering Strategy:** This is known as "behind-the-meter" generation. The data center operates effectively as an independent electrical island, powered from day one by dozens of on-site gas combustion turbines. By generating their own power locally, they completely bypass the years-long utility interconnection queue, trading long-term zero-carbon goals for immediate operational dominance in the AI market.

4 The Thermodynamics of Co-Location (Cogeneration)

Data centers do not just consume electricity; they convert it almost entirely into low-grade heat, requiring massive cooling infrastructure. Co-locating a nuclear power plant directly with a data center allows for **Combined Heat and Power (CHP)**.

Rather than venting residual steam from a turbine to a cooling tower or a local river, this low-quality heat can be captured to drive *absorption chillers*. This creates a perfect synergy: the reactor provides electricity to power the server racks, while its waste heat drives the chillers to cool them, drastically slashing the parasitic electrical demand of traditional HVAC systems.

However, there is a strict thermodynamic constraint on which reactors can effectively perform this role:

- **The Temperature Floor for Chillers:** To effectively drive a standard single-effect Lithium Bromide (LiBr) chiller, the heat source must be roughly 90°C to 120°C. High-efficiency *double-effect* chillers require temperatures upwards of 150°C to 170°C.
- **Why LWRs Struggle:** A traditional Light Water Reactor (PWR/BWR) maximizes its Rankine cycle efficiency by expanding steam until it condenses at very low temperatures ($\approx 30°C$ to $40°C$). To acquire 150°C steam from an LWR, operators must bleed high-pressure steam directly out of the turbine early, imposing a massive penalty on electrical generation.
- **The Advanced Reactor Advantage:** High-temperature Generation IV reactors (HTGRs, LMFRs, and FHRs) operate with primary core temperatures between 650°C and 850°C. Because their thermodynamic ceiling is so much higher, their power cycles can be designed to reject waste heat at 150°C—or extract it with a proportionally much smaller electrical penalty than an LWR. This thermal profile makes high-temperature advanced reactors the ideal pairings for gigawatt-scale data center complexes.

How Absorption Chillers Work: The "Thermal Compressor"

Traditional vapor-compression air conditioning uses a power-hungry mechanical pump to compress a refrigerant. An absorption chiller replaces this mechanical compressor with a *thermal* one, utilizing a **Lithium Bromide (LiBr)** and **Water** solution operating in a near-vacuum (≈ 0.01 atm).

1. The Shared Vacuum Shell (Evaporator & Absorber)
To keep the liquid water and the liquid LiBr separated while allowing vapor to transfer between them, they are housed in distinct zones within the same low-pressure steel shell.

- *Evaporator (Cooling the Data Center):* Liquid water is sprayed over a bundle of tubes carrying the data center's warm **Chilled Water Loop**. Because of the extreme vacuum, the sprayed water flash-boils at just 4°C, absorbing heat from the tube bundle to send ice-cold water back to the server racks.

- *Absorber (The Chemical "Pull" & Waste Heat):* In the adjacent zone, concentrated liquid LiBr is sprayed over a second set of tubes. As the highly hygroscopic LiBr falls, it absorbs the water vapor drifting over from the evaporator. Because this chemical absorption is highly *exothermic*, the LiBr would quickly overheat and stop absorbing if left alone. Therefore, the tubes it falls over belong to the **Cooling Tower Loop**, which constantly strips this waste heat away and rejects it to the outside environment to keep the LiBr cool and "thirsty."

2. The Generator & Condenser (The Heat Input)
The now-dilute LiBr solution falls into a sump and is pumped out of the vacuum shell to the **Generator**.

- *Generator:* This is where the reactor's heat (at roughly 150°C) is applied. The thermal energy boils the water *out* of the dilute LiBr. The hot, re-concentrated LiBr passes through a **Solution Heat Exchanger**—which cools it down while simultaneously preheating the incoming dilute solution—before it is pumped back to the Absorber to grab more vapor.

- *Condenser:* The liberated water vapor flows to a condenser, also rejects its heat to the outside environment, turns back into a liquid, and returns to the Evaporator to repeat the cooling cycle.

3. Protecting the Vacuum (The Purge Unit)
While phase changes naturally sustain the vacuum, microscopic air leaks and internal corrosion (which produces hydrogen gas) introduce *non-condensable gases* over time. Because air and hydrogen will not condense or absorb, they threaten to destroy the vacuum. Industrial chillers use an automated mechanical **Purge Unit** to continuously siphon off these stray gases and protect the thermodynamic cycle.

5 SMRs: The Natural Successor

For long-term, zero-carbon cogeneration, Small Modular Reactors (SMRs) are uniquely suited to AI co-location.

- **Lower Water Usage:** Traditional gigawatt Light Water Reactors (LWRs) require massive heat sinks (Lake Michigan or the Ohio River).
- **Geographic Flexibility:** Advanced high-temperature SMRs (like HTGRs) can utilize dry air cooling or drastically reduced water volumes, allowing them to be sited directly on landlocked data center campuses in places like Granger or New Carlisle.

6 The Legislative Fast-Track (SEA 258)

Recognizing the economic stakes, the Indiana state government has proactively removed regulatory friction.

- **Senate Enrolled Act 258 (2026):** This landmark legislation significantly limits the ability of the Indiana Department of Environmental Management (IDEM) to require additional state-level permits for nuclear facilities.
- **Overriding NIMBY:** By deferring entirely to the federal NRC, the state substantially reduces the ability of local jurisdictions to block projects once federal approval is granted.

7 Corporate Matchmaking: Who Gets What?

The hyperscalers are actively funding the next generation of nuclear technology. Based on current partnerships, we can forecast the likely reactor deployments:

- **Amazon (New Carlisle and Hobart):** AWS is heavily invested in **X-energy**. Expect proposals for the Xe-100 High-Temperature Gas Reactor (HTGR).
- **Microsoft (Granger):** Partnered closely with **Constellation Energy** and Bill Gates's **TerraPower** (Sodium-cooled fast reactors).
- **Google (Fort Wayne):** Formally partnered with **Kairos Power** to deploy fluoride salt-cooled high-temperature reactors (KP-FHR).

Engineering Reality Check: Will this happen by 2030?

The short answer is: No. While the economic incentives and legislative pathways are clear, a decade is an eternity in both politics and technology. The hyperscaler plans assume that the "First-of-a-Kind" (FOAK) prototypes currently being built (like the Xe-100 at Dow's Seadrift plant in Texas) will operate flawlessly, and the earliest will not come on line until 2028. It will take a year or so of experience for them to provide the necessary experimental evidence to satisfy the DOE/NRC MOU and could then achieve expedited approval. At that point, however, SMR construction may occur rapidly.

8 Thermal Pollution: The "Data Heat Island" Effect

While much of the public and regulatory discourse focuses on electrical grid capacity, the thermodynamic reality of computing is that 100% of the electrical energy consumed by a server rack is ultimately converted into low-grade heat. When combined with localized, fossil-fueled power generation, this creates an unprecedented thermal footprint.

8.1 The New Carlisle Thermal Load

To understand the local impact, we can calculate the continuous thermal rejection of the combined Amazon (AWS) and St. Joseph Energy Center site using the First Law of Thermodynamics.

- **The Data Center (AWS):** Consumes an estimated 2.25 GW of electricity. This entire electrical load is dissipated as heat, requiring 2.25 GW of continuous thermal rejection to the atmosphere via cooling towers, chillers, or exhaust fans.
- **The Power Plant (St. Joseph CCGT):** Generates roughly 1.4 GW of electrical power. Assuming a standard combined-cycle thermal efficiency (η_{th}) of 50%, the plant requires 2.8 GW of thermal input from natural gas combustion. The remaining 50% (1.4 GW) is rejected as waste heat.

Total Local Heat Rejection: 2.25 GW+1.4 GW = **3.65 GW of continuous thermal energy** vented directly into the local microclimate of St. Joseph County upwind of Notre Dame.

8.2 Microclimate Warming

A recent 2026 study out of the University of Cambridge (not yet peer reviewed) quantified this phenomenon, formally identifying the **"Data Heat Island" effect**. By analyzing satellite telemetry of global hyperscale deployments, researchers found severe localized warming:

- **Average Warming:** Land surface temperatures (LST) increase by an average of 2°C (3.6°F) following the commissioning of a major AI data center.
- **Radius of Impact:** This thermal dome extends significantly outward, with measurable warming observed up to 10 kilometers (6.2 miles) downwind of the facility.
- **Extreme Scenarios:** In specific geographic microclimates with high cluster density and heat-retaining infrastructure, localized temperature spikes of up to 9.1°C (16°F) have been recorded.

For communities situated immediately downwind of a 3.65 GW combined thermal source like the New Carlisle site, the engineering challenge extends beyond the electrical grid to fundamental environmental thermodynamics. This continuous heat rejection may alter local microclimate weather patterns, increase summertime HVAC loads for surrounding residents, and potentially impact local agricultural evaporation rates.

8.3 Contextualizing 3.65 GW of Thermal Rejection

To help visualize the sheer magnitude of 3.65 GW of continuous heat being dumped into a localized microclimate, we can compare it to three engineering and environmental benchmarks:

- **Comparison 1: A Commercial Nuclear Reactor Core**
  A standard 4-loop Pressurized Water Reactor (PWR), like those at the Braidwood Generating Station in neighboring Illinois, has a total core thermal power rating of approximately **3.4 GWt** (Gigawatts-thermal). Because a nuclear plant converts roughly 33% of that heat into electricity, it only rejects about 2.2 GWt to its cooling lake. The combined Amazon/St. Joseph site will reject 3.65 GWt directly into the local environment. This is the thermodynamic equivalent of venting the *entire thermal output of a large commercial nuclear core*

straight into the air, or rejecting the waste heat of nearly *two* gigawatt-scale nuclear power plants.

- **Comparison 2: High-Noon Solar Irradiance**
  The solar constant (the power of the sun hitting the Earth's surface on a clear day at high noon) is roughly $1,000$ W/m^2 (1 kW/m^2). To passively generate 3.65 GW of thermal energy, you would need to perfectly absorb the high-noon summer sun over an area of **3.65 square kilometers** (about 900 acres or 680 football fields). The data center complex generates this peak-summer solar heat load constantly—24 hours a day, 365 days a year, even in the dead of winter.

- **Comparison 3: The Anthropogenic Heat of a City**
  Urban environments generate localized heat through car exhaust, human metabolism, building HVAC systems, and industrial activity. Research into urban thermodynamics shows that the total anthropogenic heat flux of a densely populated city like **Indianapolis** (population $\sim 880,000$) averages around 2 to 3 GW continuously. The New Carlisle campus is comparable to the entire metabolic and mechanical heat output of Indiana's largest city, concentrated onto a single parcel of land.

8.4 Estimated Downwind Temperature Rise at Notre Dame

We consider a hypothetical colocated data center + power plant with total heat rejection $P = 3.65\times 10^9$ W located at the St Joseph Energy Center (approximately 15–20 km west of the University of Notre Dame). The existing facility is a $\sim$700 MW$_e$ combined-cycle plant, and we assume additional expansion and colocated data-center heat rejection brings the total thermal discharge to the multi-gigawatt scale.

Atmospheric Assumptions:

- Wind speed: $v = 5\,\mathrm{m/s}$ (typical synoptic conditions)
- Effective plume width at Notre Dame: $W \sim$ 3,000–5,000 m (due to lateral spreading over 15–20 km)
- Effective vertical mixing: enhanced by plume rise and entrainment, giving $M''_{\text{eff}} \approx$ 2000–4000 kg/m^2

Mass Flow Estimate:

$$\begin{aligned}\dot m &\approx W \cdot v \cdot M''_{\text{eff}} \\ &\sim (3\times10^3 \text{ to } 5\times10^3)\cdot 5\cdot(2\times10^3 \text{ to } 4\times10^3) \\ &\sim (3 \text{ to } 10)\times 10^7\,\mathrm{kg/s}\end{aligned}$$

Resulting Temperature Rise:

$$\begin{aligned}\Delta T &= \frac{P}{\dot m c_p} \\ &\approx \frac{3.65\times10^9}{(3\text{–}10)\times10^7\cdot 1005} \\ &\approx \mathbf{0.1\text{–}0.4}^\circ\mathrm{C} \quad (\mathbf{0.2\text{–}0.7}^\circ\mathrm{F})\end{aligned}$$

Interpretation: At the distance of Notre Dame, atmospheric mixing, plume rise, and entrainment substantially dilute the thermal discharge. Although the specific heat of air is not exceptionally large, the sheer mass of air moving through the control volume (set by wind speed and atmospheric depth) is enormous, strongly limiting the resulting temperature rise. Under typical daytime conditions, the expected increase in bulk air temperature is on the order of a few tenths of a degree Celsius.

Larger transient increases (approaching $\sim 1°$F) may occur under stagnant or weakly mixed atmospheric conditions, but sustained multi-degree temperature increases at this distance are not likely. Closer to the source, however - particularly within the first few kilometers downwind - reduced plume dilution and incomplete vertical mixing can produce temperature increases several times larger than those observed at Notre Dame.

Surface (land) temperature increases near large heat sources can exceed several degrees Celsius due to localized heat flux and low thermal mass. However, these effects are spatially confined. In contrast, air temperature increases are smaller in magnitude but are transported and diluted over long distances by atmospheric mixing.

9 The Economic Fallout: Ratepayer Impact in Northern Indiana

The deployment of hyperscale data centers does not just alter the physical grid; it fundamentally rewires the regional energy economy. To understand what will happen to utility bills in Northern Indiana, we must first look at the precedent set on the East Coast, and then map those mechanics onto our local utility jurisdictions.

9.1 The PJM Warning: The Virginia/New Jersey Case Study

The eastern half of the United States operates on a massive, synchronized power grid managed by the PJM Interconnection. PJM ensures grid reliability by running an annual "Capacity Auction." This auction does not pay for electricity itself; rather, it acts as an insurance policy, paying power plants simply to exist and be ready to dispatch power during peak demand events.

Historically, this insurance was cheap. However, as Virginia's "Data Center Alley" rapidly consumed the grid's excess reserve capacity, the supply of backup power plummeted. As a result, the clearing price for capacity in PJM has risen dramatically in recent auctions from roughly \$28 per megawatt-day, to in some cases approaching the market cap of over \$330/MW-day.

Because PJM socializes these capacity costs across all 13 states in its territory, residents in New Jersey saw their residential electric bills spike by approximately 20%. They are effectively subsidizing the grid insurance required to support data centers located hundreds of miles away in Virginia.

9.2 The Local Electric Split: I&M vs. NIPSCO

Northern Indiana is uniquely positioned on the seam of two different grid operators. Consequently, the economic impact of local data centers may differ depending on which side of the county line you live on.

- **The East (South Bend/Granger – I&M / PJM):** In a counter-intuitive short-term twist, customers of Indiana Michigan Power (I&M) are currently seeing a direct financial benefit. Because tech giants like Amazon (which is building a massive 2.2 GW campus in New Carlisle), Microsoft, and Google are committing to massive, guaranteed power purchases, I&M announced in February 2026 that it will file to reduce residential base rates using this influx of corporate revenue. However, because I&M operates within the PJM interconnection, this area remains vulnerable to future capacity market spikes if regional generation fails to keep pace with the tech growth.

- **The West (Hobart/Gary – NIPSCO / MISO):** Just to our west, the grid shifts to the Midcontinent Independent System Operator (MISO), serviced by NIPSCO. Here, ratepayers recently absorbed a significant base rate hike (finalized in March 2026) to fund grid infrastructure upgrades. In an attempt to soften the blow, NIPSCO announced in April 2026 that it will issue direct bill credits—estimated by the utility to be around \$90 to \$115 annually per household—using early revenue from its new GenCo subsidiary agreements with Amazon and Google. However, with massive new facilities like the highly contested Hobart AWS data center coming online, it remains uncertain how long these credits will offset the staggering infrastructure costs required to deliver gigawatts of new power.

9.3 The Universal Hit: Natural Gas Commodities

While the electric bill impacts are geographically split, the impact on natural gas bills will likely be universally felt. Because renewable energy cannot currently provide the unyielding, 24/7 baseload power required by data centers, utilities in both MISO and PJM are relying heavily on natural gas turbines to firm the grid. NIPSCO provides natural gas to the vast majority of Northern Indiana.

These gigawatt-scale tech campuses effectively act as continuous, massive consumers of natural gas, creating intense regional competition for pipeline capacity. As utilities and on-site ”behind-the-meter” facilities draw vast quantities of gas to generate electricity, local residential consumers will face permanent upward pressure on both the delivery and commodity costs of natural gas. Ultimately, this creates the very real risk of structural increases in winter heating bills across the entire Michiana region.

9.4 Funding the Nuclear Future: Who Pays for SMRs?

To ultimately transition these gigawatt-scale data centers off natural gas, the energy sector is pursuing advanced Small Modular Reactors (SMRs). Because these localized projects will be commercial deployments rather than early federal demonstration prototypes, the Department of Energy is no longer footing the bill. Instead, the financial risk is handled in two entirely different ways, depending on who builds the reactor.

Path 1: The Regulated Utility Model (Public Risk)

When a traditional public utility builds an SMR for the grid—such as AEP's proposed BWRX-300 at the Rockport site—the financial burden and risk fall heavily on the local ratepayer.

- **Construction Work In Progress (CWIP):** In 2022, the Indiana legislature passed Senate Bill 271, legally allowing regulated public utilities to use CWIP accounting for SMRs.
- **The Ratepayer Reality:** CWIP means utilities do not have to wait until the nuclear plant is finished to start charging for it. **Ratepayers will see a monthly surcharge on their utility bills to finance the construction years before the reactor ever generates a single watt of electricity.** If the project goes over budget or faces delays, the local ratepayers absorb the financial risk, not the utility's shareholders.

Path 2: The Off-Grid Hyperscaler Model (Private Risk)

Because tech giants cannot wait a decade for utilities to navigate public financing, future SMR projects located near data center hubs in Northern Indiana will likely utilize an “off-grid” or “behind-the-meter” model.

- **Private Capital & IPPs:** In this model, an Independent Power Producer (IPP) partners with the tech giant to build the reactor exclusively to power the campus.
- **Corporate PPAs:** The massive, trillion-dollar balance sheets of hyperscalers (Amazon, Google, Microsoft) are used to absorb the commercial risk. They fund these projects through direct equity or multi-decade Power Purchase Agreements (PPAs) that guarantee the developer a buyer for the electricity.
- **The Consumer Shield:** Because these are private, unregulated power plants not selling to the public grid, CWIP does not apply. If the reactor goes over budget or falls behind schedule, the tech shareholders and private equity backers take the loss. **The local residential ratepayer pays nothing for its construction or cost overruns.**

10 Where Do We Go From Here? Extrapolating the AI Energy Curve

The 8.35 GW projection for Northern Indiana represents a monumental engineering challenge, but from a computational scaling perspective it may even underestimate future demand. We are currently experiencing a digital manifestation of the **Jevons Paradox**: as hardware becomes more electrically efficient, the algorithmic demand for computation is growing exponentially faster than the efficiency gains.

Current AI scaling laws indicate that to achieve artificial general intelligence (AGI), training clusters will need to scale up by orders of magnitude. We are rapidly approaching a reality where a single training cluster may require multiple gigawatts—exceeding the capacity of any single commercial SMR or even traditional gigawatt-scale reactor. If terrestrial grids and local microclimates cannot sustain tens of gigawatts of continuous thermal rejection, the tech industry must look for a different heat sink.

Speculative Engineering: Off-World AI and the Radiation Hurdle

The Orbital Data Center

To bypass the terrestrial constraints of grid queues, freshwater consumption, and the ”Data Heat Island” effect, tech visionaries and aerospace startups are increasingly proposing space-based data centers. The thermodynamic logic is appealing:

- **Infinite Energy:** Infrastructure in High Earth Orbit (HEO) or on the lunar surface has access to unattenuated, near-continuous solar irradiance (1.36 kW/m^2), completely decoupling the data center from the terrestrial baseload.
- **Infinite Heat Sink:** While the vacuum of space prevents convective cooling, it offers an infinite thermal sink for radiative cooling (T^4 via the Stefan-Boltzmann Law). Massive deployable radiators can reject thermal energy continuously without impacting a local biosphere.

The Nuclear Engineering Roadblock: Radiation

The ultimate barrier to putting artificial intelligence in space is not launch mass or latency; it is the ionizing radiation environment. This brings the AI pipeline directly into the realm of nuclear engineering physics.

- **Single Event Upsets (SEUs):** Modern AI chips (GPUs/TPUs) are built on sub-3nm architectures operating at fractions of a volt. A single galactic cosmic ray, solar proton, or trapped electron traversing the silicon will deposit enough ionization energy to flip a bit (an SEU) or trigger a catastrophic short circuit (latch-up). Hallucinations in terrestrial AI are software issues; in space, they are hardware-induced by ionizing radiation.

- **The Shielding Mass Penalty:** To run cutting-edge, commercial GPUs in orbit, they must be heavily shielded. Because high-energy particle and Bremsstrahlung shielding requires significant atomic mass (lead, water, or high-density polymers), the required shielding weight severely penalizes the orbital launch economics via the Tsiolkovsky rocket equation.

- **The Rad-Hardened Performance Gap:** The aerospace industry builds radiation-hardened chips (e.g., using Silicon-on-Insulator technology), but because of the physics of rad-hardening, these chips are physically larger, vastly slower, and mathematically generations behind commercial AI silicon.

Takeaway: Until materials science engineers fundamentally new, lightweight radiation-shielding metamaterials—or computer scientists develop entirely new, radiation-fault-tolerant computing architectures—the AI revolution remains inextricably tethered to the Earth's surface, and by extension, to the terrestrial baseload grid.

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